Precept 5: Warm Up Exercises

October 7, 2026

Exercises

Problem 1

Let $\vec{u} = \begin{bmatrix}1\\2\end{bmatrix}$ and $\vec{v} = \begin{bmatrix}3\\-1\end{bmatrix}$, and define $A = \vec{u}\vec{v}^\top$.

(a) Compute A and $\det A$.

(b) What is the rank of A? How many non-zero singular values does it have?

(c) Without computing the full SVD, what is the condition number of A?

(d) Find a non-zero vector in the null space of A.

Problem 2

Someone hands you the SVD $A = USV^\top$, where

$$U = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & -1\\1 & 1\end{bmatrix}, \qquad S = \begin{bmatrix}4 & 0\\0 & \frac{1}{2}\end{bmatrix}, \qquad V^\top = \begin{bmatrix}0 & 1\\1 & 0\end{bmatrix}$$

(a) Compute $\det U$ and $\det V^\top$. What kinds of transformations are U and $V^\top$? Rotations or reflections?

(b) Compute $\det A$ **without** multiplying out A.

(c) A unit square is transformed by A. What is the area of the resulting shape?

(d) Compute $\det\left(A^{-1}\right)$ and $\det\left(A^\top A\right)$.

(e) What is the condition number of A?

Solution

Problem 1

Subproblem 1.a

A is an outer product, and we can think of the outer product as

$$A = \begin{bmatrix} | & | \\ v_1\vec{u} & v_2\vec{u} \\ | & | \end{bmatrix}$$

where each column is the vector $\vec{u}$ scalar multiplied by the corresponding component of the vector $\vec{v}$. Hence, the matrix A is

$$A = \begin{bmatrix} 3 & -1 \\ 6 & -2 \end{bmatrix}$$

Remember that the formula for the determinant of a 2×2 matrix is

$$\det A = ad - bc$$

where

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

Plugging our particular components of $A = \vec{u}\vec{v}^\top$, we have

$$\det A = 3\,(-2) - (-1)\,6 = -6 + 6 = 0$$

Subproblem 1.b

As we've seen before, the rank of a matrix formed by an outer product is just 1. This follows because there is one linearly independent vector in the column space of A, namely the vector $\vec{u}$.

Our matrix A has 1 non-zero singular value. You should ask yourself, "Why is this?" Let's think back to the SVD. We can write the SVD as

$$A = USV^\top$$

We know that U, V are orthogonal matrices[1]; hence, the rank of U, V are both rank 2. But we know that the rank of A is 1, so where does the reduction in the rank of the SVD come from? It comes from S!

In fact, the rank of A always equals the rank of S in the SVD. **You should convince yourself of this.** Now we know that the rank of S is equal to 1. Given that S is a diagonal matrix, this implies that:

- one of the columns has a non-zero vector
- the other column is the zero vector $\vec{0}$

Therefore, there is one non-zero singular value and one zero singular value.

Thinking back to **Subproblem 1.a**, we computed the determinant of A as 0. In class, we saw that the determinant of A is also the product of all the singular values. Given we have one non-zero and one zero singular value, their product is equal to 0, giving us a nice sanity check!

Subproblem 1.c

Following from our logic in **Subproblem 1.b**, A has two singular values:

1. A non-zero singular values σ_1
2. A zero singular value $\sigma_2 = 0$

[1]Just as a reminder, all the columns of an orthogonal matrix are linearly independent, hence the rank of an orthogonal matrix is always equal to the number of columns. In this case, U, V are 2×2 matrices, so there are two columns, implying they are rank 2.

We compute the condition number as the largest singular value divided by the smallest singular value, which in our case is

$$\frac{\sigma_1}{\sigma_2} = \frac{\sigma_1}{0} = \infty$$

Notice that it does not matter what σ_1 is equal to!

Subproblem 1.d

Given that we know the rank of A is 1, we can infer that the dimension of the null space is 1, hence there are vectors that live there. To find them, let's suppose we have some arbitrary vector $\vec{x} = \begin{bmatrix} x_1 & x_2 \end{bmatrix}^\top$ that we want to lie in the null space. Now we can rewrite the matrix multiplication $A\vec{x} = 0$ as

$$A\vec{x} = \vec{u}\vec{v}^\top\vec{x} = 0$$

where we can see that $\vec{v}^\top\vec{x} = 0$ implies $A\vec{x} = 0$. Therefore, we just need to find a vector orthogonal to $\vec{v}$. Explicitly writing out the components of $\vec{v}^\top\vec{x} = 0$, we get

$$\begin{aligned} \vec{v}^\top\vec{x} &= 0 \\ \begin{bmatrix} 3 & -1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= 0 \\ 3x_1 - x_2 &= 0 \\ x_2 &= 3x_1 \end{aligned}$$

Now, all we have to do is to pick an arbitrary value of x_1. Let's say $x_1 = 1$. Via our expression above, it follows that $x_2 = 3$. Therefore, our vector $\vec{x}$ is

$$\vec{x} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

and, by construction, it lies in the null space. Do double check!

Problem 2

Subproblem 2.a

Using our formula for the determinant

$$\det A = ad - bc$$

we have

$$\det U = \left(\frac{1}{\sqrt{2}}\right)\cdot\left(\frac{1}{\sqrt{2}}\right) - \left(-\frac{1}{\sqrt{2}}\right)\cdot\left(\frac{1}{\sqrt{2}}\right) = \frac{1}{2} + \frac{1}{2} = 1$$

and

$$\det V^\top = 0\cdot 0 - 1\cdot 1 = -1$$

A $\det U = 1$ implies that U is a rotation and a $\det V^\top = -1$ implies $V^\top$ is a reflection.

Subproblem 2.b

From class, we know that

$$\begin{aligned}\det A &= \det\left(USV^\top\right) \\ &= \det U \cdot \det S \cdot \det V^\top\end{aligned}$$

We already know $\det U$ and $\det V^\top$, so we just need to figure out $\det S$. Luckily, S is a diagonal matrix, so the determinant of S is the product of its diagonal components

$$\det S = 4 \cdot \frac{1}{2} = 2$$

Combining everything, we have

$$\det A = \det U \cdot \det S \cdot \det V^\top = 1 \cdot 2 \cdot (-1) = -2$$

Subproblem 2.c

Recall from class that the determinant of a matrix is a measure of how much the volume is increased or decreased by applying that matrix to a shape[2]. For this problem, the shape is the unit square with area 1. Upon applying our matrix A to the square, the area is increased by $|\det A| = 2$. Hence, the area of the resulting shape is $2 \cdot 1 = 2$.

Subproblem 2.d

Recall from class that A^{-1} can be computed from the SVD as

$$A^{-1} = VS^{-1}U^\top$$

Hence,

$$\det\left(A^{-1}\right) = \det V \cdot \det S^{-1} \cdot \det U^\top$$

You should convince yourself that

$$\det V = \det V^\top \qquad \text{and} \qquad \det U = \det U^\top$$

Now all we need to do is to compute $\det S^{-1}$, which is just

$$\det S^{-1} = \frac{1}{4} \cdot \frac{1}{1/2} = \frac{1}{4} \cdot 2 = \frac{1}{2}$$

Putting it all together, we have

$$\det\left(A^{-1}\right) = \det V \cdot \det S^{-1} \cdot \det U^\top = (-1) \cdot \frac{1}{2} \cdot 1 = -\frac{1}{2}$$

Notice that we miraculously have

$$\det\left(A^{-1}\right) = \frac{1}{\det A}$$

which is always true for an invertible matrix A. Pretty cool, right?

[2]It's okay if the idea of multiplying a shape by a matrix is unclear. Everyone in the linear algebra world becomes comfortable with this idea by watching this 3blue1brown video.

Recall from class, and the quiz, that $A^\top A$ can be written via the SVD of A as

$$A^\top A = VS^2V^\top$$

It now follows that we can write the determinant of $A^\top A$ as

$$\begin{aligned}\det\left(A^\top A\right) &= \det\left(VS^2V^\top\right) \\ &= \det V \cdot \det S \cdot \det S \cdot \det V^\top \\ &= -1 \cdot 2 \cdot 2 \cdot (-1) = 4\end{aligned}$$

Notice that we have another miracle. We can write

$$\det\left(A^\top A\right) = (\det A)^2$$

which is always true for any square matrix A. Even cooler, right?

Subproblem 2.e

The condition number is just the largest singular value divided by the smallest singular value. Hence, it is

$$\frac{4}{1/2} = 8$$